Efficient DiffWave Audio Inpainting With Resampling

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Background

- **Audio inpainting** involves the reconstruction of missing audio. There is a wide range of applications, like the removal of glitches from recordings.
- DiffWave is an efficient denoising diffusion probabilistic model (DDPM) used for raw audio synthesis. It is not equipped to perform audio inpainting [1].
- How can we enhance DiffWave to perform audio inpainting?
- We take inspiration from RePaint, which provides an approach for using a pretrained, unconditional DDPM for the task of image inpainting [2].
- Specifically, we implement RePaint's **inpainting algorithm** with **resampling**.
- As input, the model accepts raw audio as a .wav file and a mask as a numpy array. It outputs the reconstructed audio as a wav file.

Challenges

- The data types for RePaint and DiffWave are different. How do we reconcile this?
- Should the parameters for our algorithm, like alpha and beta, change because of the data?
- At what point does resampling saturate and begin to become inefficient?

Dataset

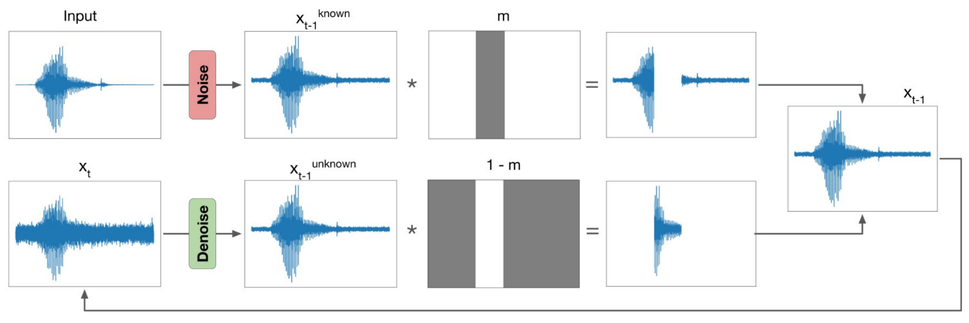
- Our dataset is the SC09 dataset, which consists of raw audio waveforms [3].
- Each clip in the dataset is a one second long utterance of a digit between 0 and 9.
- The speakers come with a diverse set of accents, pitches, and other noise conditions.

Method

- We perform audio inpainting with resampling under the following algorithm:

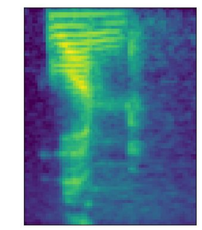
$x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, \ldots, 1$ **do**
  for $u = 1, \ldots, U$ **do**
    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\epsilon = \mathbf{0}$
    $x_{t-1}^{\text{known}} = \sqrt{\bar{\alpha}_t} x_0 + (1 - \bar{\alpha}_t)\epsilon$
    $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
    $x_{t-1}^{\text{unknown}} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(x_t, t) \right) + \sigma_t z$
    $x_{t-1} = m \odot x_{t-1}^{\text{known}} + (1 - m) \odot x_{t-1}^{\text{unknown}}$
    if $u < U$ and $t > 1$ **then**
      $x_t \sim \mathcal{N}(\sqrt{1 - \beta_{t-1}} x_{t-1}, \beta_{t-1}\mathbf{I})$

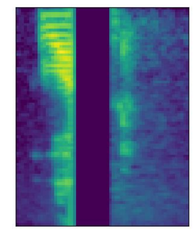
- Note that α bar is the total noise variance, m is the mask, ε_θ is the VAE for the predicted noise, and α/β are constants.
- We perform T forward/reverse steps, resampling U times per step.
- We compute the output of the **known** and **unknown** regions by denoising. We then **combine our guesses** for the audio in both regions.
- This process is illustrated below.



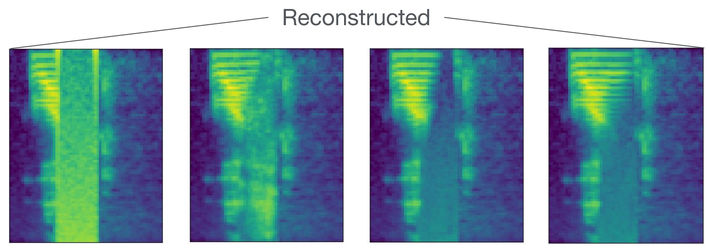
Evaluation

- We demonstrate the ability of our adapted DiffWave model on inpainted audio for corrupted audio files on the digit 1 for U = 1, 2, 4 (live demo)


Original


Masked

Reconstructed


Random U=1 U=2 U=4

Reconstruction Method	MSE with original audio
Random	0.0197
DiffWave U=1	0.0089
DiffWave U=2	0.0078
DiffWave U=4	0.0034

References

- [1] Kong, Z., Ping, W., Huang, J., Zhao, K., & Catanzaro, B. (2021, March 30). *DiffWave: A versatile diffusion model for audio synthesis*. arXiv.org. https://arxiv.org/abs/2009.09761
- [2] Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., & Van Gool, L. (2022a, August 31). *Repaint: Inpainting using denoising diffusion probabilistic models*. arXiv.org. https://arxiv.org/abs/2201.09865
- [3] Goel, K. (n.d.). *SC09*. krandiash/sc09 · Datasets at Hugging Face. https://huggingface.co/datasets/krandiash/sc09